

Theorem (Markov's Inequality). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Assume $X : \Omega \rightarrow [0, \infty]$ is a non-negative random variable and let $a > 0$. Then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. 1. **Trivial cases.** If $\mathbb{E}[X] = +\infty$ the right-hand side is $+\infty$ and the inequality is automatically true. If $\mathbb{P}(X \geq a) = 0$ the inequality reads $0 \leq \mathbb{E}[X]/a$, which holds because $\mathbb{E}[X] \geq 0$. Hence we may assume $0 < \mathbb{P}(X \geq a) < 1$ and $\mathbb{E}[X] < \infty$.

2. **Indicator bound.** Define the indicator of the event $\{X \geq a\}$ by

$$I_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

For each $\omega \in \Omega$,

$$X(\omega) \geq a I_{\{X \geq a\}}(\omega). \quad (2.1)$$

Verification of (2.1): if $X(\omega) \geq a$ then $I_{\{X \geq a\}}(\omega) = 1$ and $a \leq X(\omega)$; if $X(\omega) < a$ then $I_{\{X \geq a\}}(\omega) = 0$ and (2.1) reduces to $X(\omega) \geq 0$, true since $X \geq 0$ a.s.

3. **Expectation of the bound.** Both sides of (2.1) are non-negative, so monotonicity of expectation yields

$$\mathbb{E}[X] \geq a \mathbb{E}[I_{\{X \geq a\}}]. \quad (3.1)$$

4. **Expectation of an indicator.** By definition of expectation for indicator functions,

$$\mathbb{E}[I_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (4.1)$$

5. **Combine (3.1) and (4.1).** Substituting (4.1) into (3.1) gives

$$\mathbb{E}[X] \geq a \mathbb{P}(X \geq a),$$

which is equivalent to

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \quad (5.1)$$

This completes the proof for finite expectations; the case $\mathbb{E}[X] = +\infty$ was settled in step 1.

6. **Equality condition.** Equality in (5.1) can occur only if equality holds both in the pointwise bound (2.1) and in the expectation monotonicity step (3.1).

- *Pointwise bound:* equality in (2.1) requires

$$X(\omega) = a I_{\{X \geq a\}}(\omega) \quad \text{for } \mathbb{P}\text{-a.e. } \omega,$$

i.e.

$$X(\omega) = \begin{cases} a, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

- *Expectation step:* if the above identity holds $\mathbb{P}$ -a.s., then taking expectations gives

$$\mathbb{E}[X] = a \mathbb{P}(X \geq a),$$

so (3.1) and consequently (5.1) become equalities.

Thus

$$\mathbb{P}(X \geq a) = \frac{\mathbb{E}[X]}{a} \iff X = a I_{\{X \geq a\}} \quad \text{a.s.},$$

i.e. X assumes only the two values 0 and a (up to a $\mathbb{P}$ -null set). In all other situations the inequality is strict.

□